The HCF of 175, 245 and 385 is the product of the common factors 5 and 7 i.e, 5 x 7=35

Since each box contains equal number of books, the greatest possible number of boxes = 35

The number of English books in each box = 175÷35=5

The number of Science books in each box = 245÷35=7

The number of Maths books in each box = 385÷35=11

Hence, the total number of books in each box is 5+7+11=23.

Note

- LCM is always greater than or equal to the largest of the given numbers.
- LCM will always be a multiple of HCF.

Example 8: Find the ratio of the HCF and the LCM of the numbers 18 and 30.

Solution: Now, 18 = 2 x 3 x 3 and 30 = 2 x 3 x 5
and their HCF is 2 x 3 = 6 and LCM is 2 x 3 x 3 x 5 = 90
Hence, HCF : LCM = 6:90 = 1:15

Example 9: Find the smallest number that can be divided by 254 and 508 which leaves the remainder 4.

Solution: All common multiples of 254 and 508 will be divisible by both the numbers. Let us find the LCM of 254 and 508 (by division method).

2	254, 508
2	127, 254
127	127, 127
	1, 1

LCM of 254, 508 = 2 x 2 x 127 = 508

Thus, 508 is the smallest common number that is divisible by 254 and 508. Now, as we need remainder 4 while dividing, the required number is 4 more than the LCM and so, the required number is 508 + 4 =512.

Example 10: What is the smallest 5 digit number that is exactly divisible by 72 and 108?

Solution: First let us find the LCM of 72 and 108 (by division method).

2	72,108
2	36,54
2	18,27
3	9,27
3	3,9
3	1,3
	1,1

LCM of 72 and 108= 2 x 2 x 2 x 3 x 3 x 3 = 216

Now, all multiples of 216 will also be common multiples of 72 and 108.

The smallest 5 digit number = 10,000.

Now, 10,000 ÷ 216 gives quotient as 46 and remainder as 164.

Hence the next multiple of 216 i.e., 216 x 47 = 10,152 is the required smallest 5 digit number that is exactly divisible by 72 and 108.

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